Weston Robot

Basic Training

A1: Robot Dog

26/05/2022

Weston Robot

Agenda

01
Main Components

02
Basic Operation

03
Advanced Operations

04
Development

Main Components

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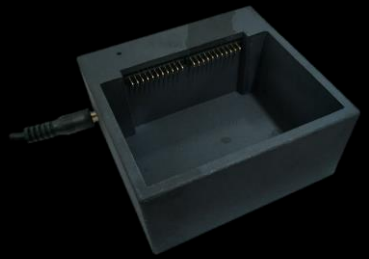
Main Components

A1

Battery

Controller

Leash



Battery Charger

Emergency Stop

Basic Operation

Switching On the Robot

1. Battery
   - Battery Level Display
   - Use provided charger and power supply
   - Overcharge Protection
   - Self-discharge Protection (65%)

1. Controller
   - Charge using USB – C
   - Tap then Hold the On button (Beep + Lights)
   - Tapping every button causes a Beep

Switching On the Robot

1. How to
   - Insert the battery
   - Tap then Hold the On button

1. Safety Precaution
   - Place A1 in Neutral Prone Position

1. Battery Precaution
   - Ensure battery level > 1 LED
   - Ensure battery was not just charged (Heating)

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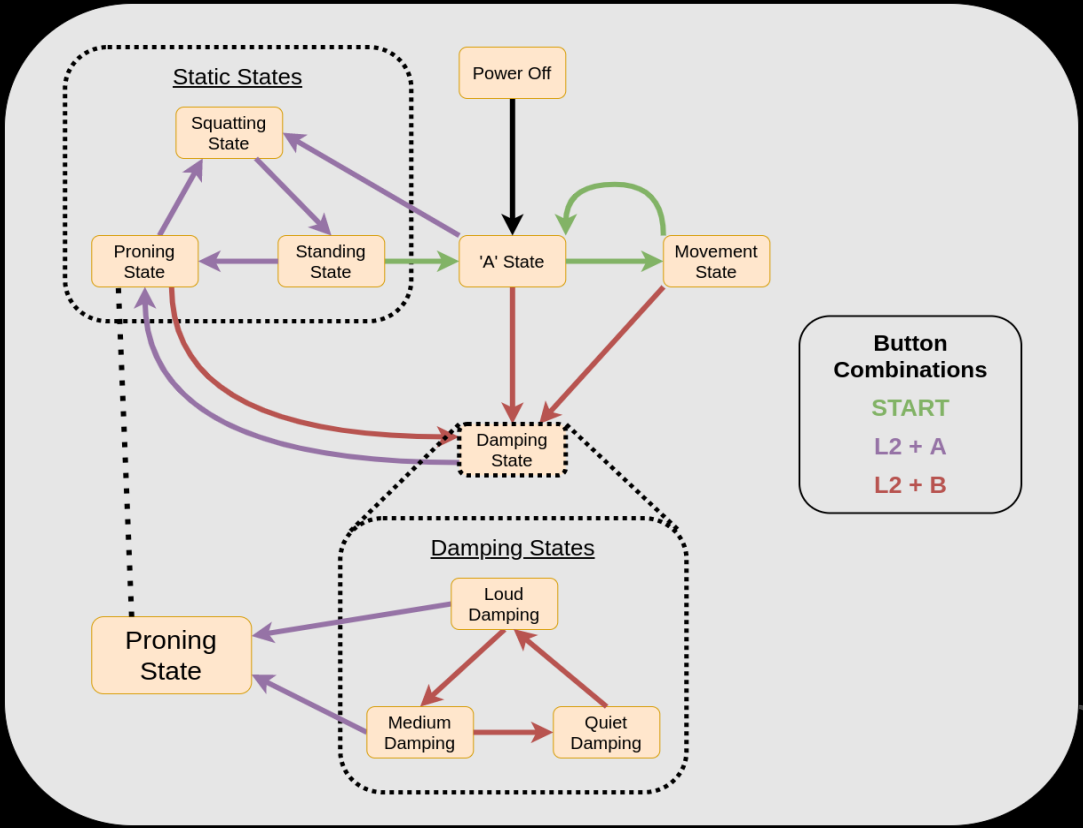
Normal Mode



Do NOT

- Slope > 22°
- Slippery / Soft Floor
- Low Battery Levels
- Use Other batteries

Basic App Usage (beta)

1. Download
   - https://www.unitree.com/app/
   - Android and IOS

1. Connect to A1: WiFi
   - UnitreeRoboticsA1-XXX
   - Password: 00000000 (Eights Zeros)
   - Ip Setting: DHCP

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Settings

Signal

APP

Dog

Video Ip	192.168.123.12
MJPEG Port	8080
H264 Port	8090
Vision Ip	192.168.123.12
Vision Port	9011
SLAM Ip	192.168.123.12
SLAM Port	5000
Controller Ip	192.168.123.12
Controller Port	8007

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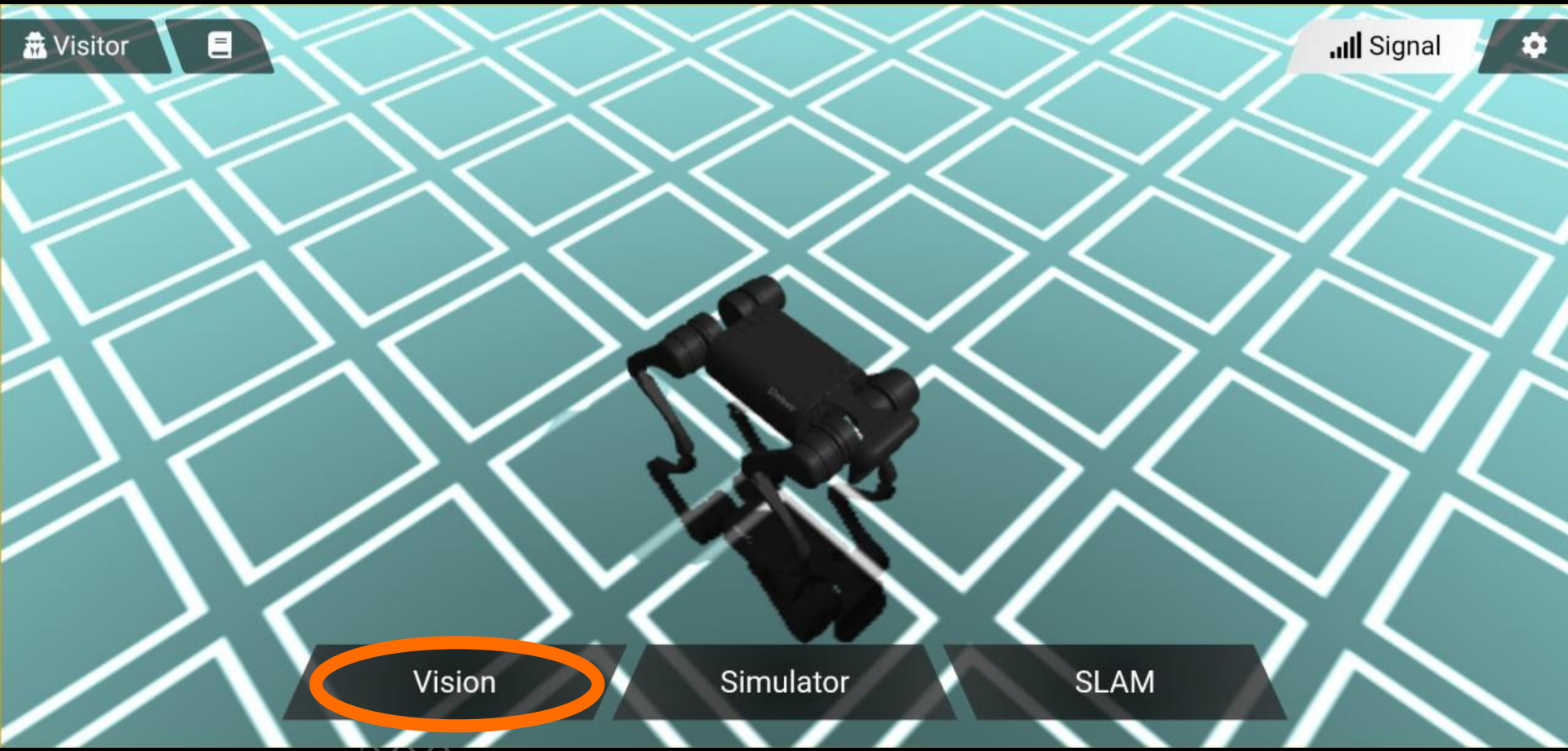

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UR

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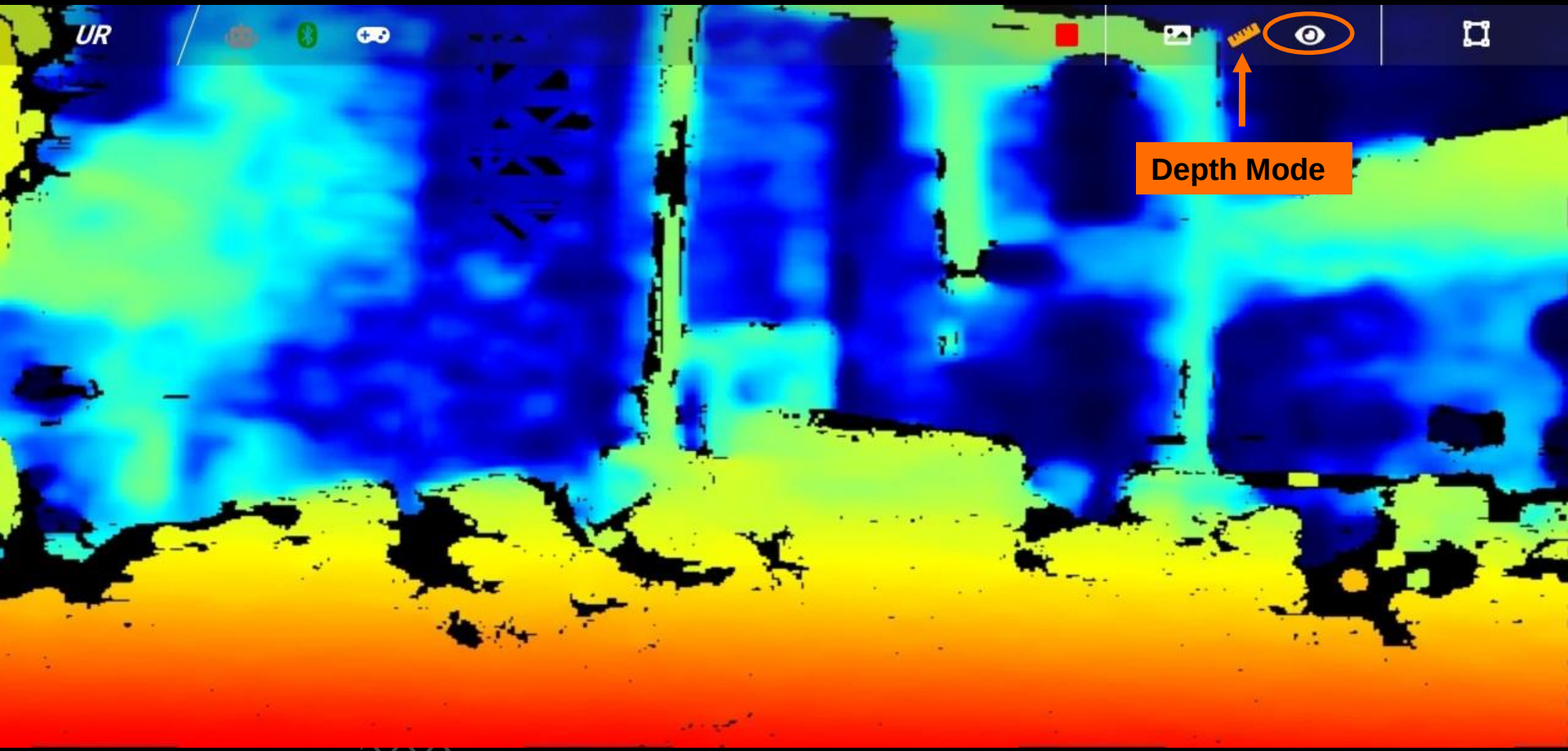
UR
Image Mode



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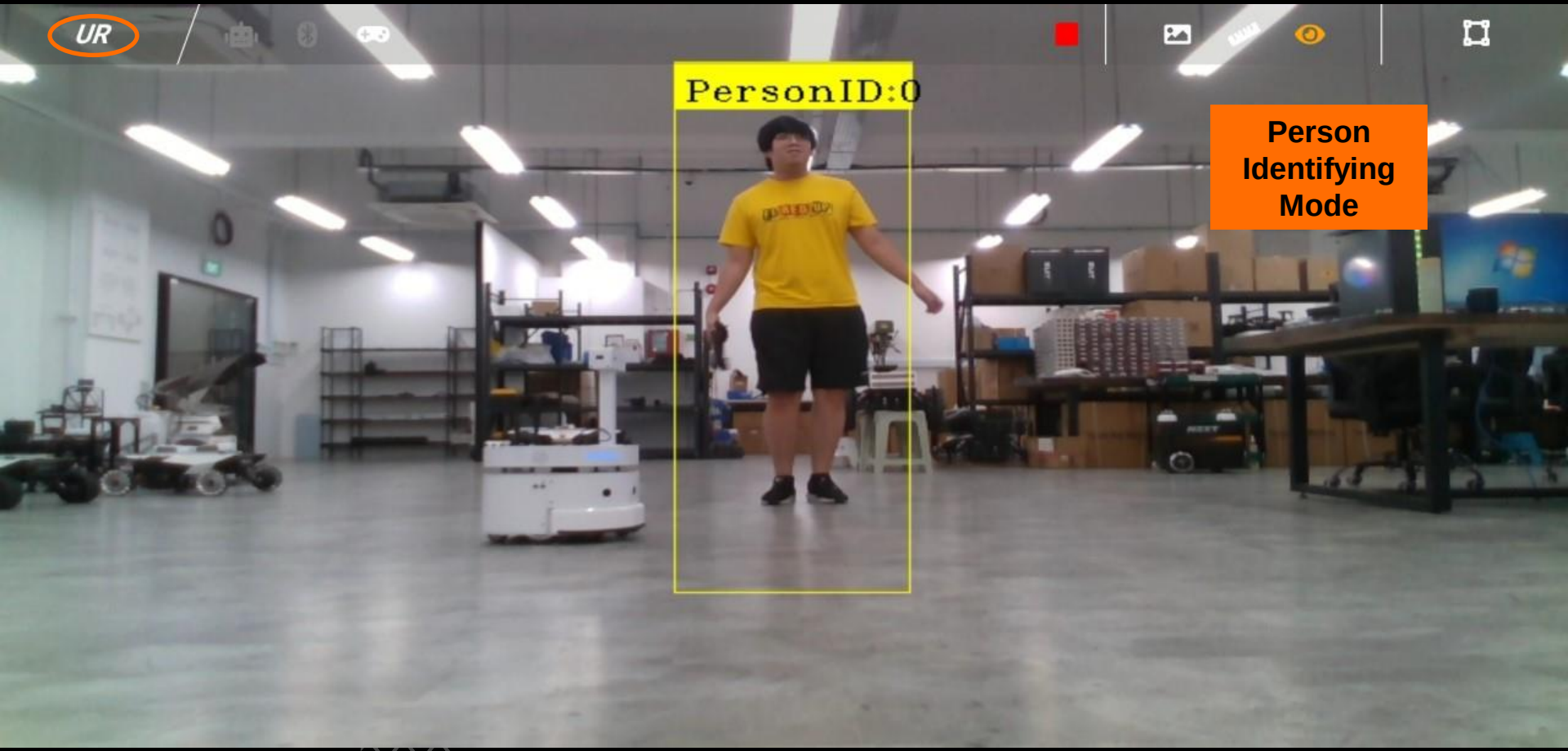
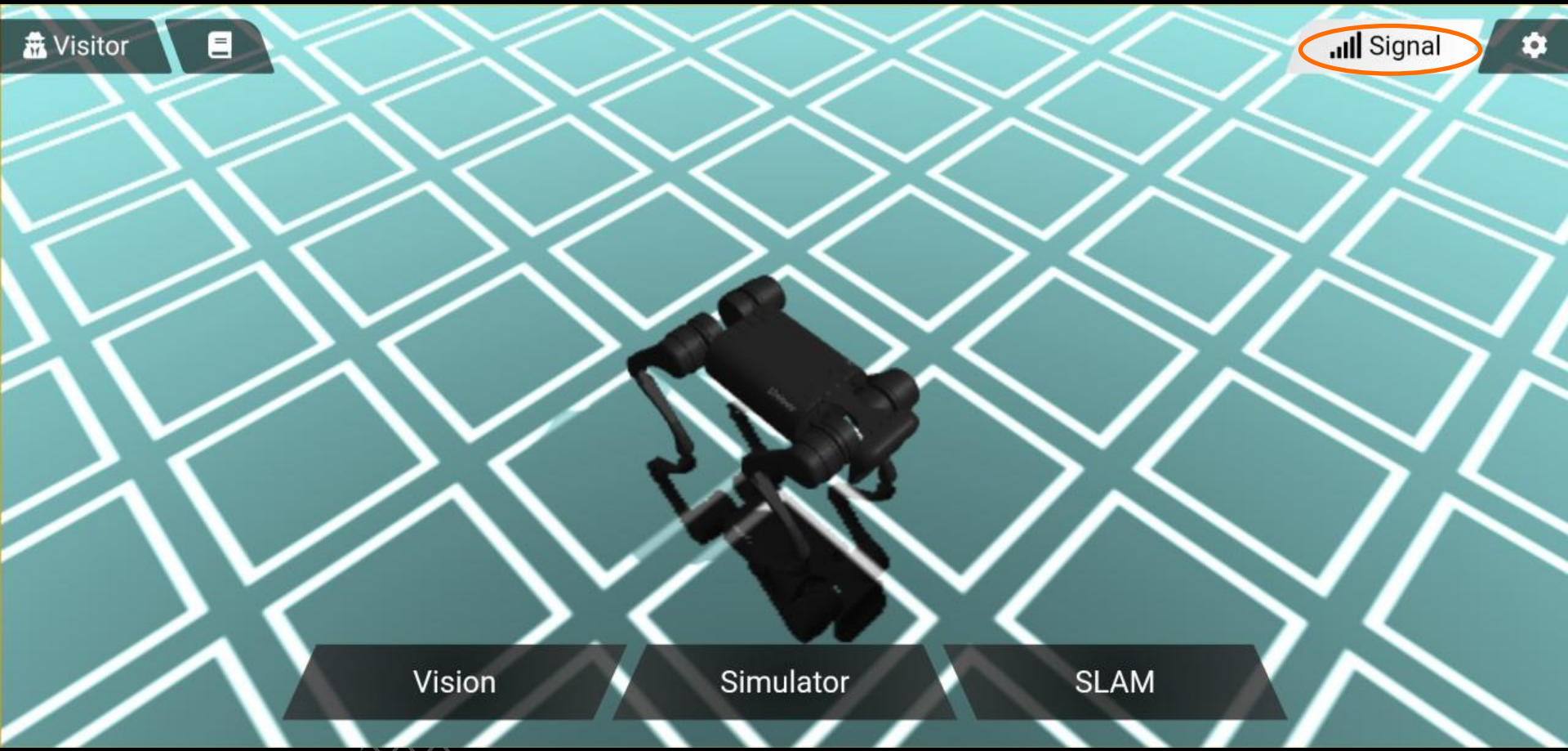


Person Identifying Mode



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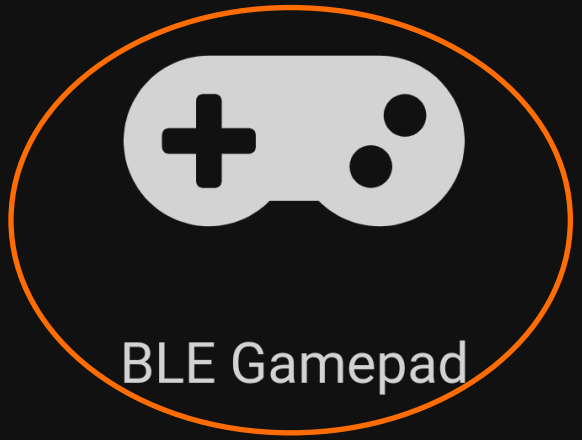
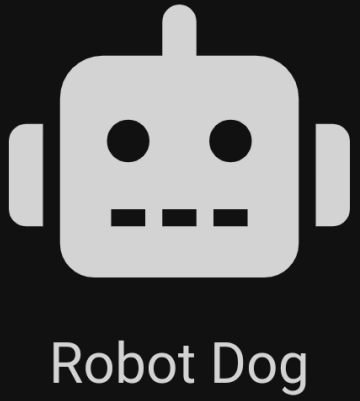
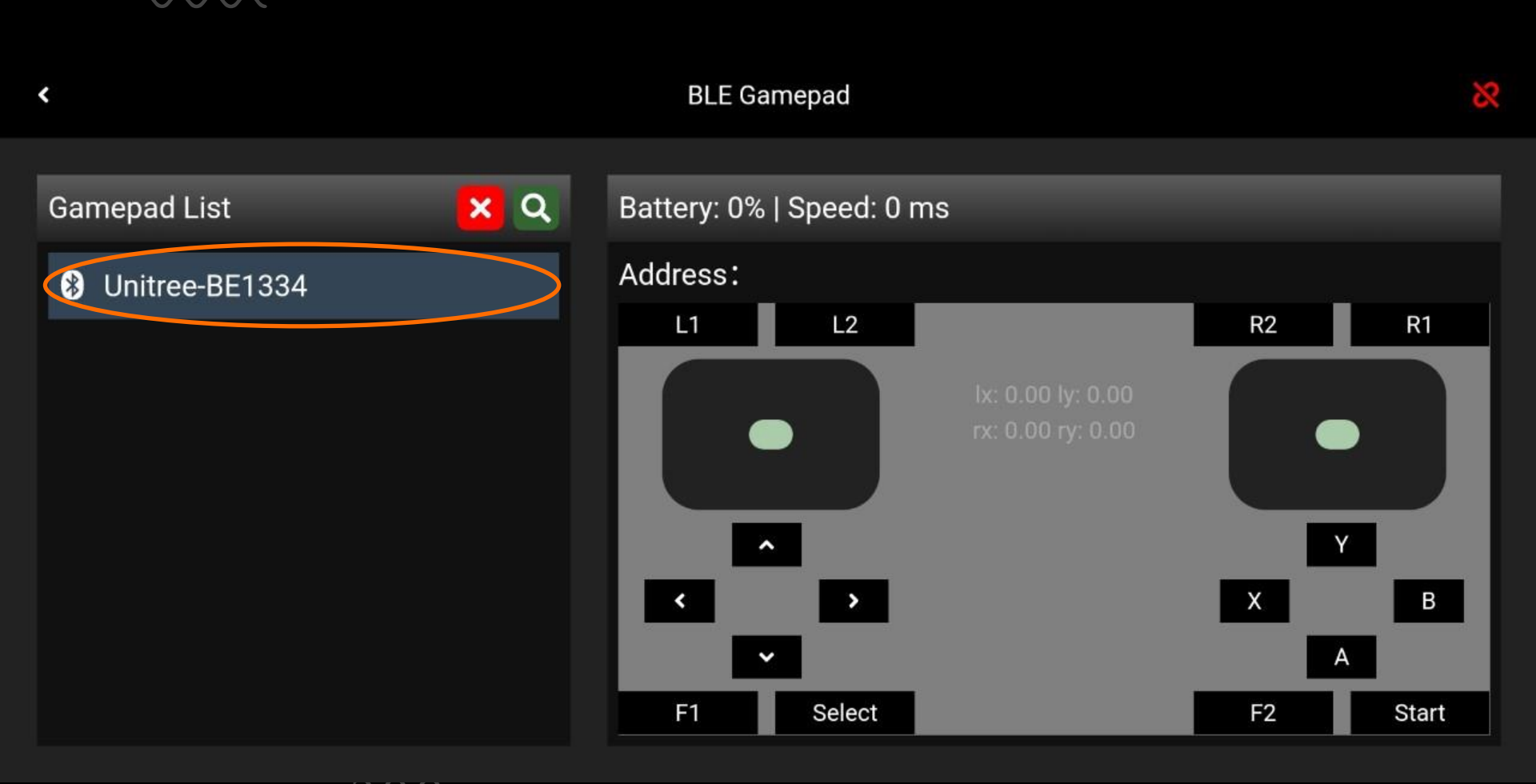
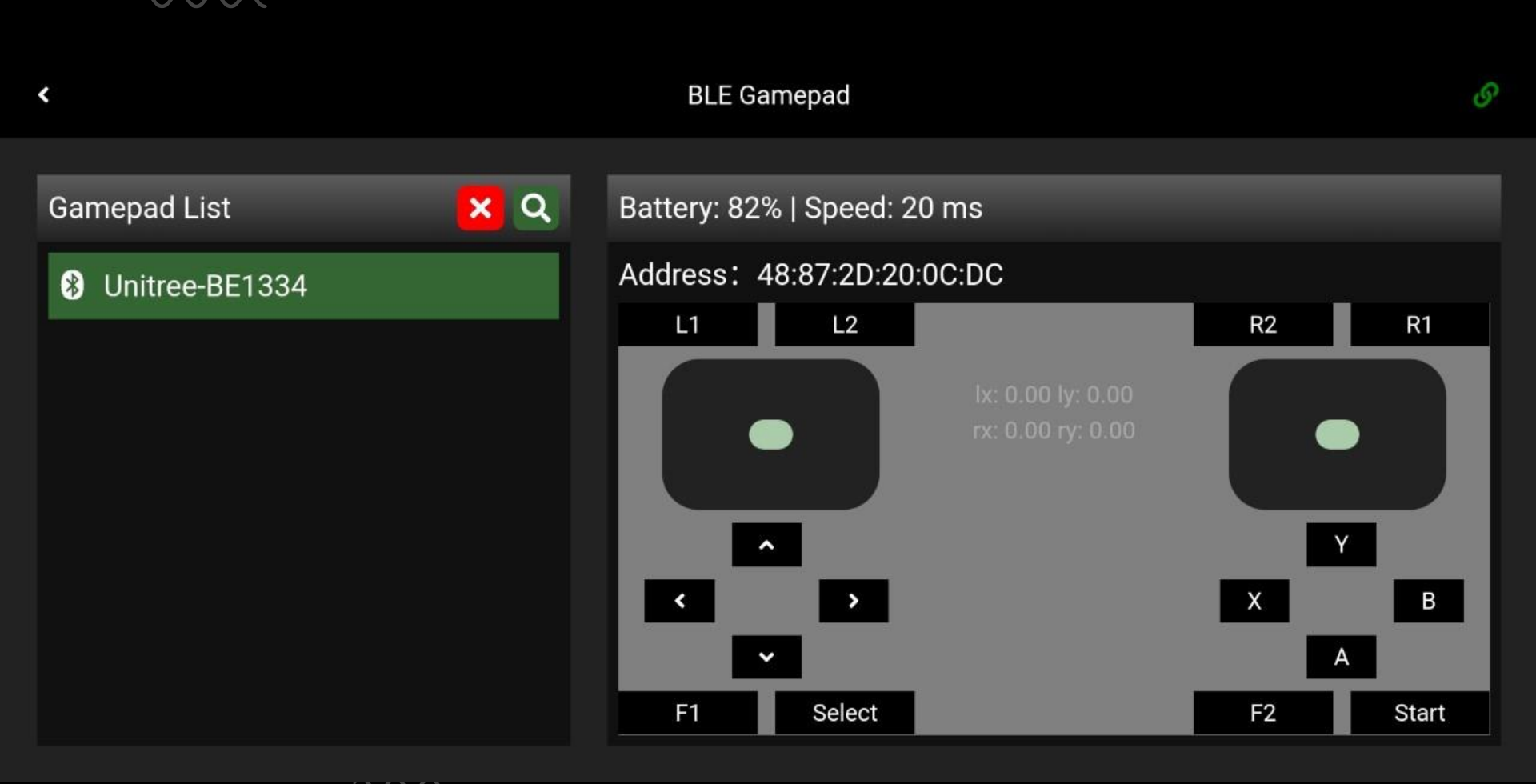
Signal Connections






BLE Gamepad
Gamepad List
Unitree-BE1334
Battery: 0% | Speed: 0 ms
Address:
L1
L2
R2
R1
lx: 0.00 ly: 0.00
rx: 0.00 ry: 0.00
Y
X
B
A
F1
Select
F2
Start


BLE Gamepad
Gamepad List
Unitree-BE1334
Battery: 82% | Speed: 20 ms
Address: 48:87:2D:20:0C:DC
L1
L2
R2
R1
lx: 0.00 ly: 0.00
rx: 0.00 ry: 0.00
Y
X
B
A
F1
Select
F2
Start

Advanced Operations

Summary of Modes in A1

	Normal Mode	Sports Mode	Sports Mode 3.0 / Climbing Mode
Static States	• 3 states (Prone, Squat, Stand)	• 2 states (Prone, Stand)	• 2 states (Prone, Stand)
Damping States	• 3 state (Quiet, Medium, Loud)	• 1 state (Quiet)	• 1 state (Quiet)
Main Advantage	• Lower power consumption	• Greater balance and stability (due to increased on-board computation) • Higher maximum speed	• Greater balance and stability (due to increased on-board computation) • Ability to perform stair-climbing (maximum of 12cm stair height)
Main Disadvantage	• Less balance • Lower maximum speed	• Higher power consumption	• Higher power consumption

Sports Mode

Overview

- Greater Balance
- Faster Speed

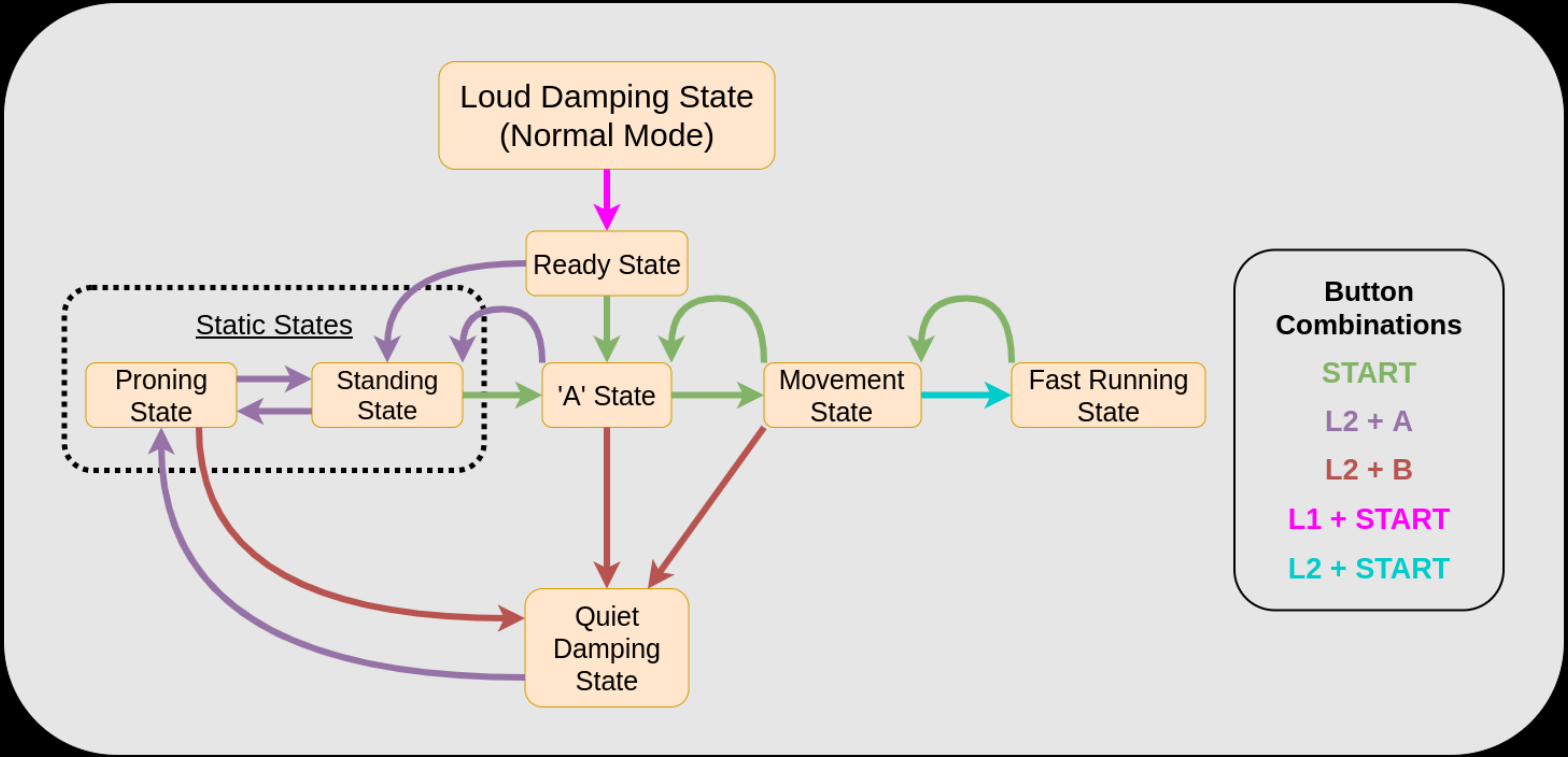
Switch to Sport Mode

- Enter Loud Damping State (Normal Mode)
- *L1 + Start*
- *Start* to enter A – State (Sports Mode)

Return to Normal Mode

- Enter Quiet Damping State (Sports Mode)
- *L2 + L1 + Start* to enter Quiet Damping State (Normal Mode)
- *L2+B* to enter Loud Damping State followed by *L2+A* to go to static state of Normal Mode

Sports Mode


Loud Damping State
(Normal Mode)
Ready State
Static States
Proning State
Standing State
'A' State
Movement State
Fast Running State
Quiet Damping State
Button Combinations
START
L2 + A
L2 + B
L1 + START
L2 + START

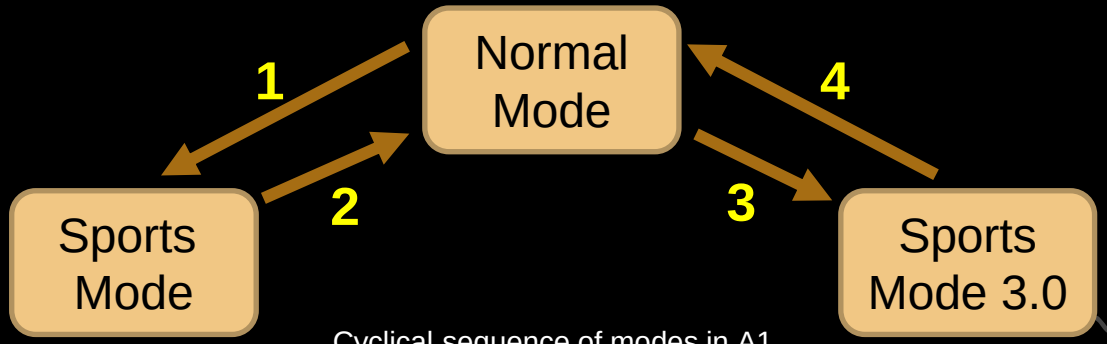
Sports Mode 3.0 / Climbing Mode

From 'A' State (Normal Mode)

- Switch to Loud Damping State (Normal Mode)
- *L1 + Start* to enter Ready State (Sports Mode) **1**
- Go to Quiet Damping State (Sports Mode)
- *L2 + L1 + Start* to return to Quiet Damping State (Normal Mode) **2 / 4**

*Note: *L2+B* to enter Loud Damping State followed by *L2+A* to go to static state of Normal Mode

- *L1 + Start* **AGAIN** to enter Standing State (Sports Mode 3.0) **3**



Cyclical sequence of modes in A1

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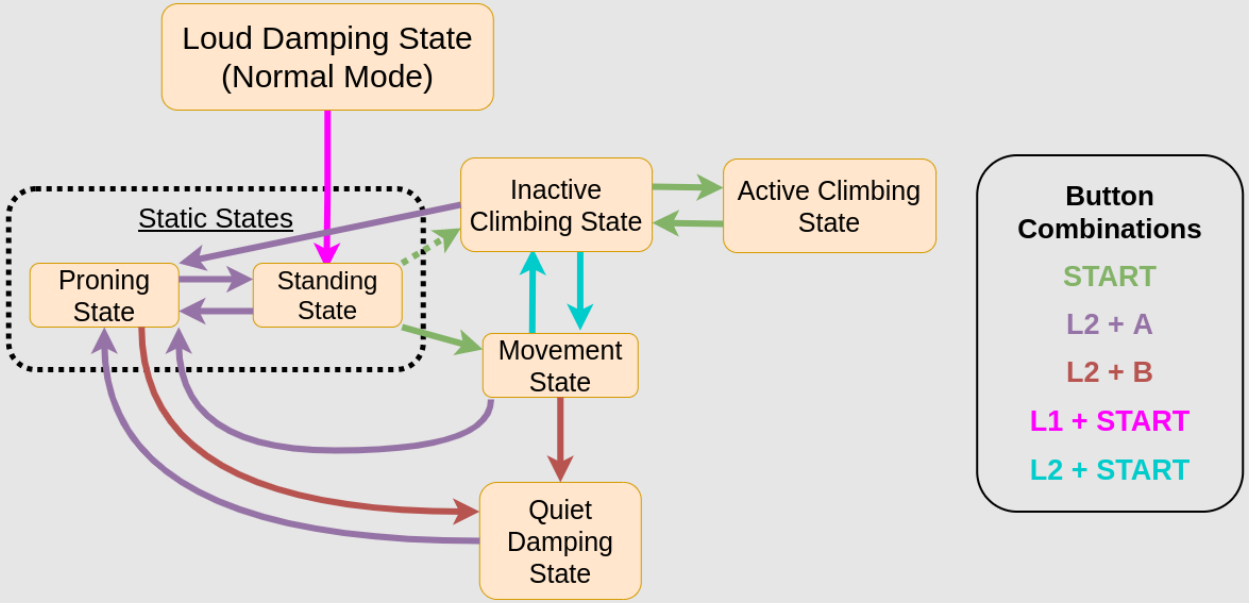
Sports Mode 3.0 / Climbing Mode



Note: By default, the A1 transitions from Standing Static state to Movement State. However, if it was last switched to Climbing State, the A1 will immediately transition from Standing Static State to Inactive Climbing State.

Summary of Modes in A1

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Main Disadvantage	• Less balance • Lower maximum speed	• Higher power consumption	• Higher power consumption

Advanced Commands

- Rolling
- Adjust Body Height
- Adjust Leg Lift
- Force Push

Rolling (360 / 180)

- Instructions
  - In A – State (Normal Mode)
  - Standing Upright / Lying Down
  - *L2 + Y* / *L2 + X*

- Pre Caution
  - Do not do this with LIDAR or other components attached
  - Ensure ample space for the flip

Adjust Body Height

- Instructions
  - In A – State (Normal Mode) / Walking State (Sports Mode)
  - *Up* / *Down Arrows*
  - *Start*

- Precaution
  - Remember number of key presses
  - Return to default setting to conserve battery
  - Carefully control speed
  - ONLY do this in walking mode

Adjust Leg Lift

- Instructions
  - In Movement State (Normal / Sports Mode)
  - *L1 + Up* / *Down Arrows*
  - *Start*

- Precaution
  - Remember number of key presses
  - Return to default setting to conserve battery
  - Higher lift, greater impact, greater wear and tear

Adjust IMU Drift

- Instructions
  - In Movement State (Normal Mode)
  - *Left* / *Right Arrows*

- Precaution
  - Wait 3 seconds after each press then observe

Following Mode

- Uses the Camera and the App
- Tracks distinct objects in bland background best
- Only operates in sports mode

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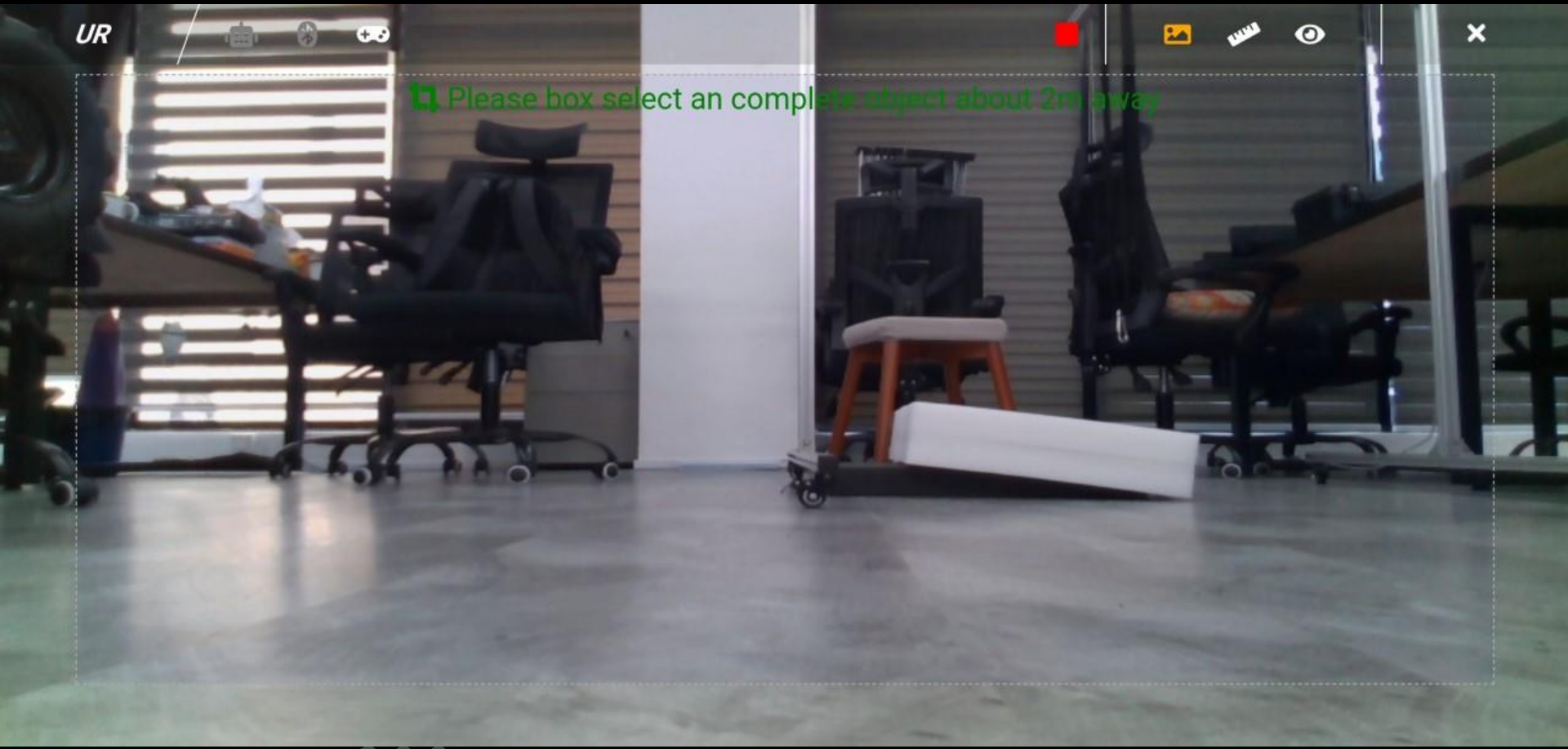
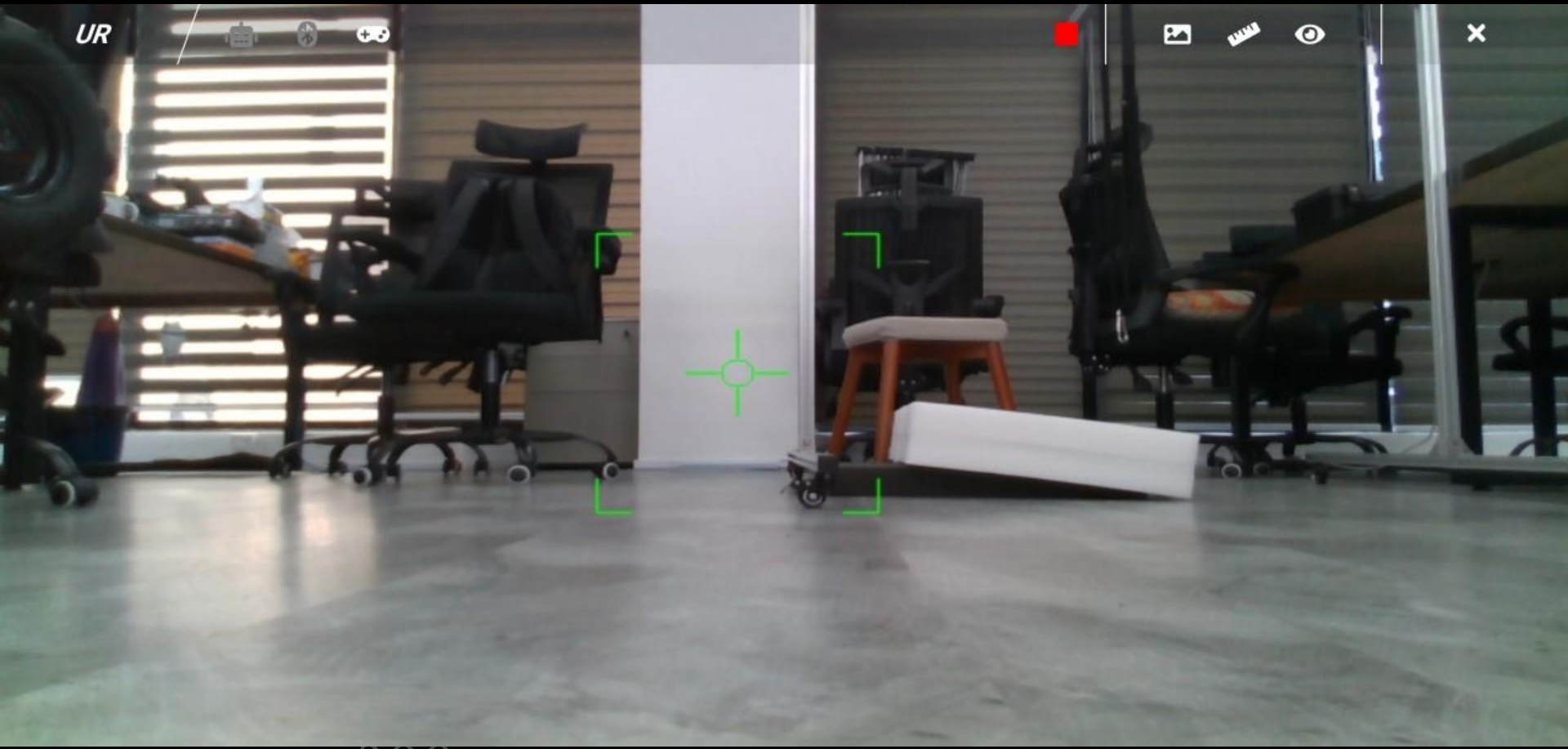

UR
Please box select an complete object about 2m away

UR
Please box select an complete object about 2m away

UR
Please box select an complete object about 2m away



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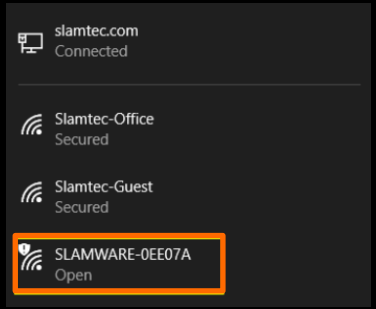
Development

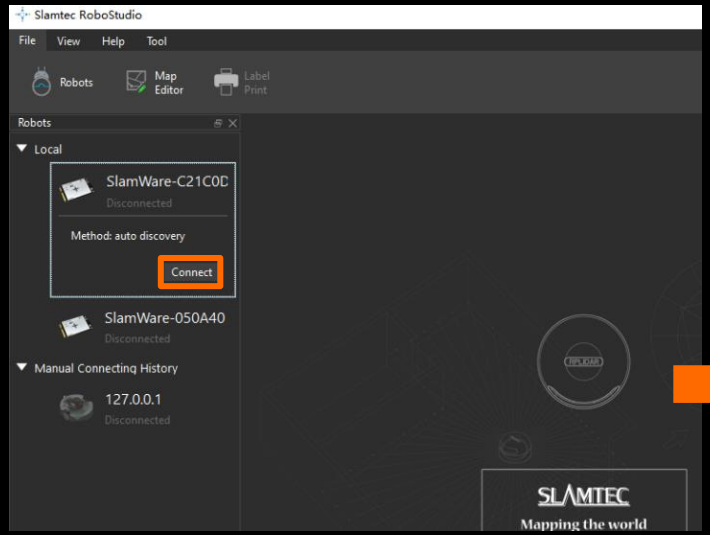
RoboStudio (2D LiDAR)

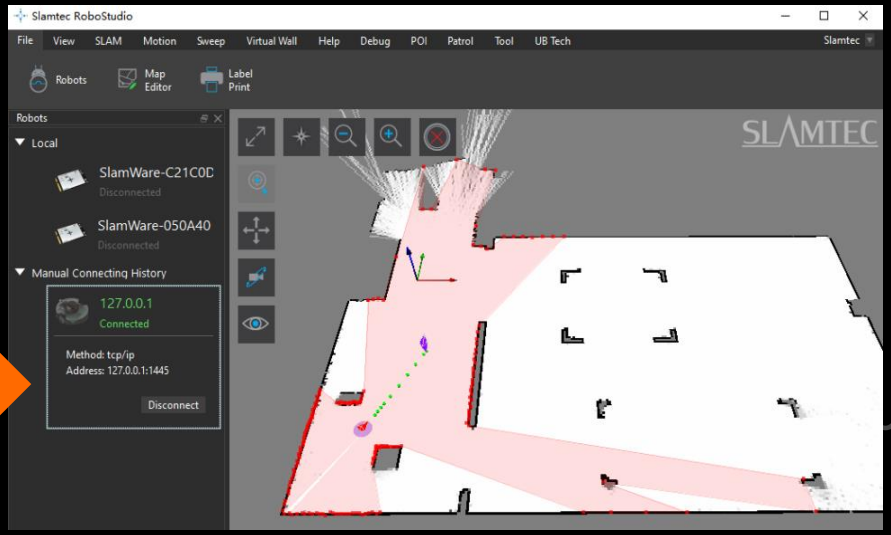
- Install RoboStudio on Windows/Android devices
  - https://www.slamtec.com/en/RoboStudio
- SLAMWARE User Manual
  - https://bucket-download.slamtec.com/e16008e591546788a026fb8c21204088ed08e24d/SA008_SLAMTEC_slamware_robot_studio_appnote_v1.1_en.pdf
- Connect to SLAMWARE Wi-Fi



2D Mapping

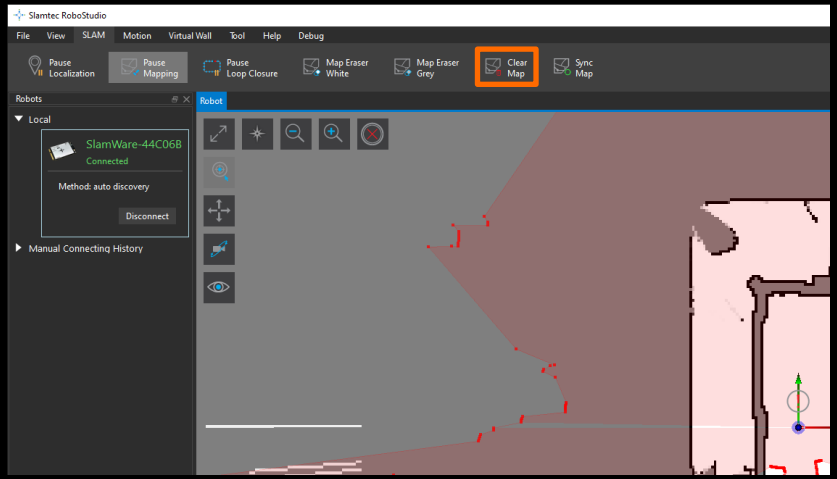
- Connect Robot





2D Mapping

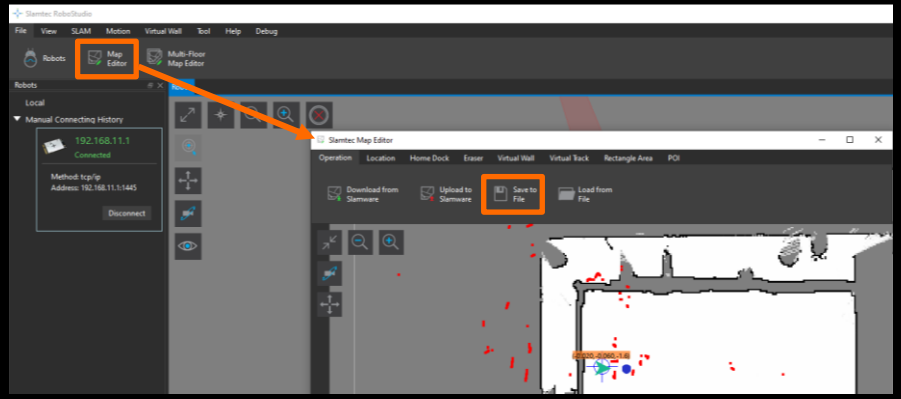
- Clear current map and start mapping



- When mapping, drive A1 in low speed and **turn extremely slowly**, otherwise robot will be delocalized which will result in inaccurate map

2D Mapping

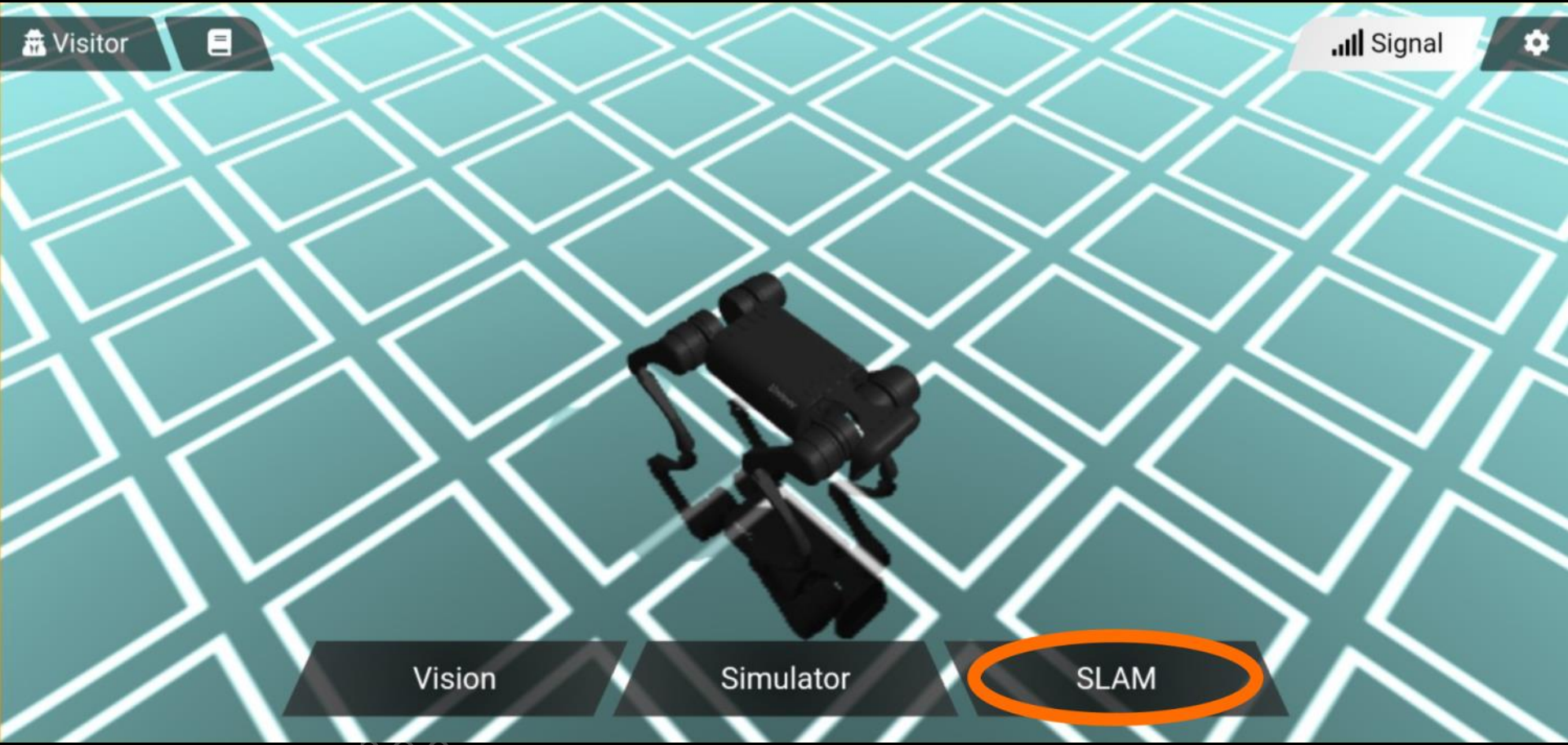
- When mapping is done, Save map by clicking on the Save to File button



- Name map as map.stcm and transfer new map file to NX board and place new map under this directory:
/home/unitree/catkin_ws/src/slamware_ros_sample/map

SLAM / Navigation Mode

- Uses the LIDAR and the App
- Camera disabled
- Only operates in sports mode



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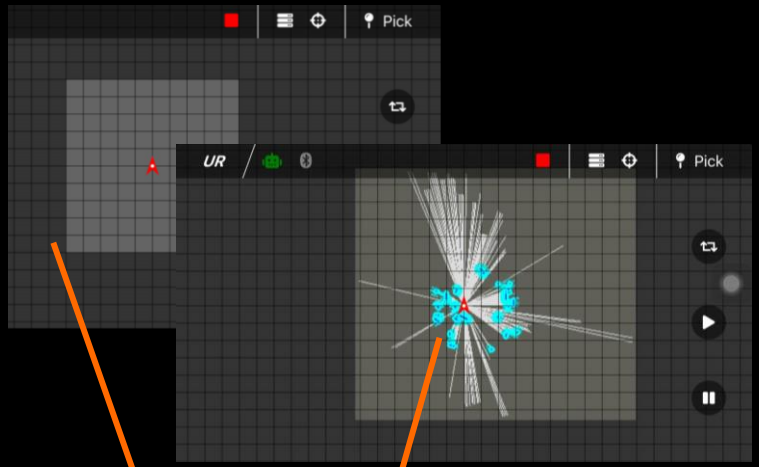
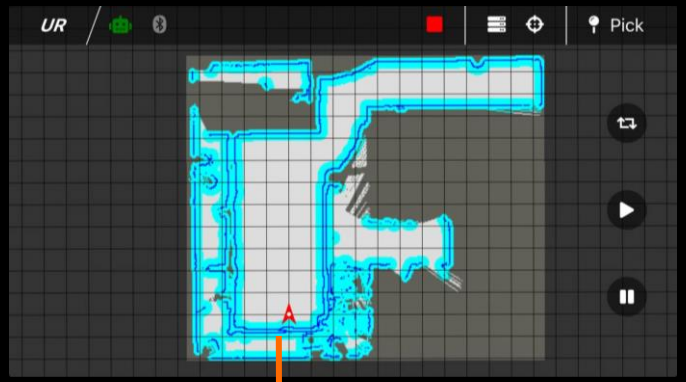
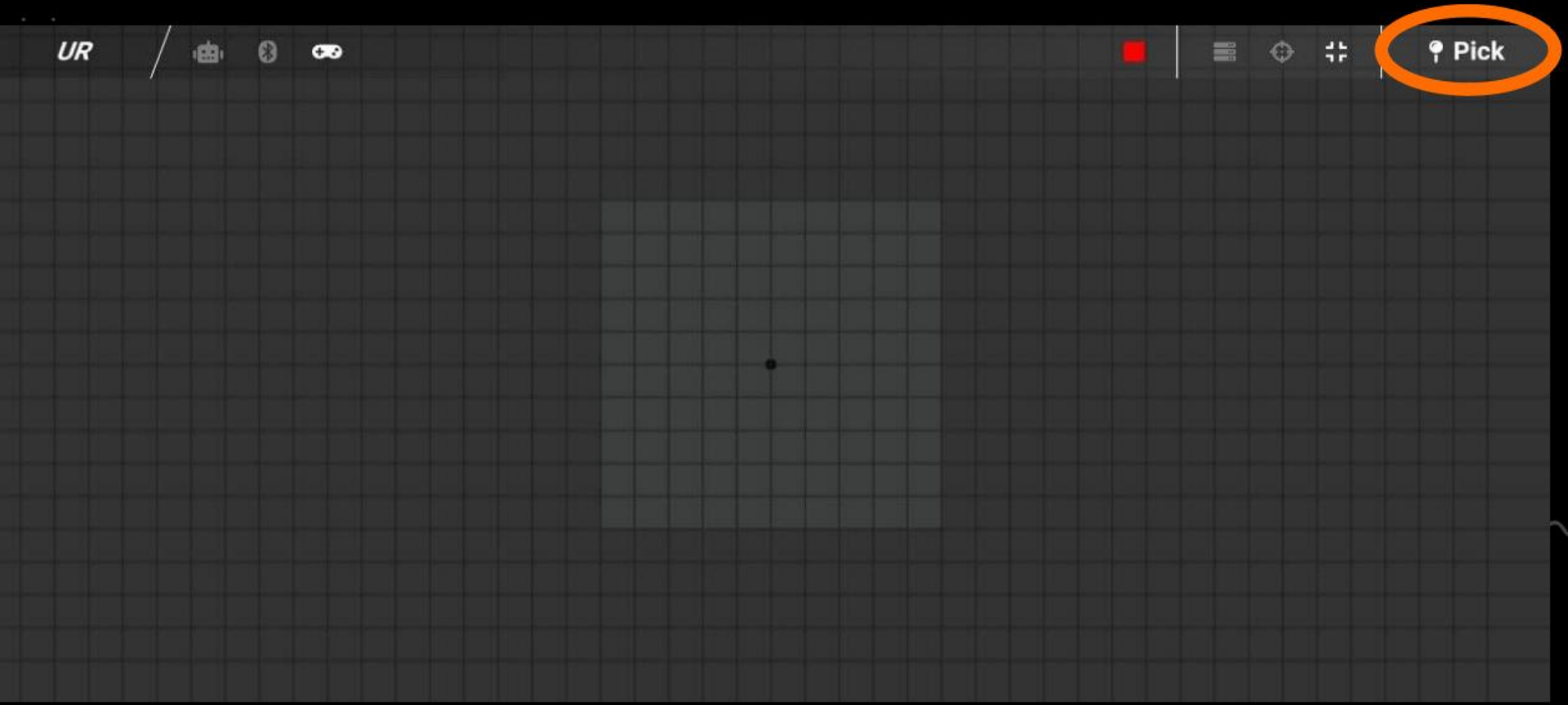
Check Map from App



No Map Loaded



Map Loaded Correctly



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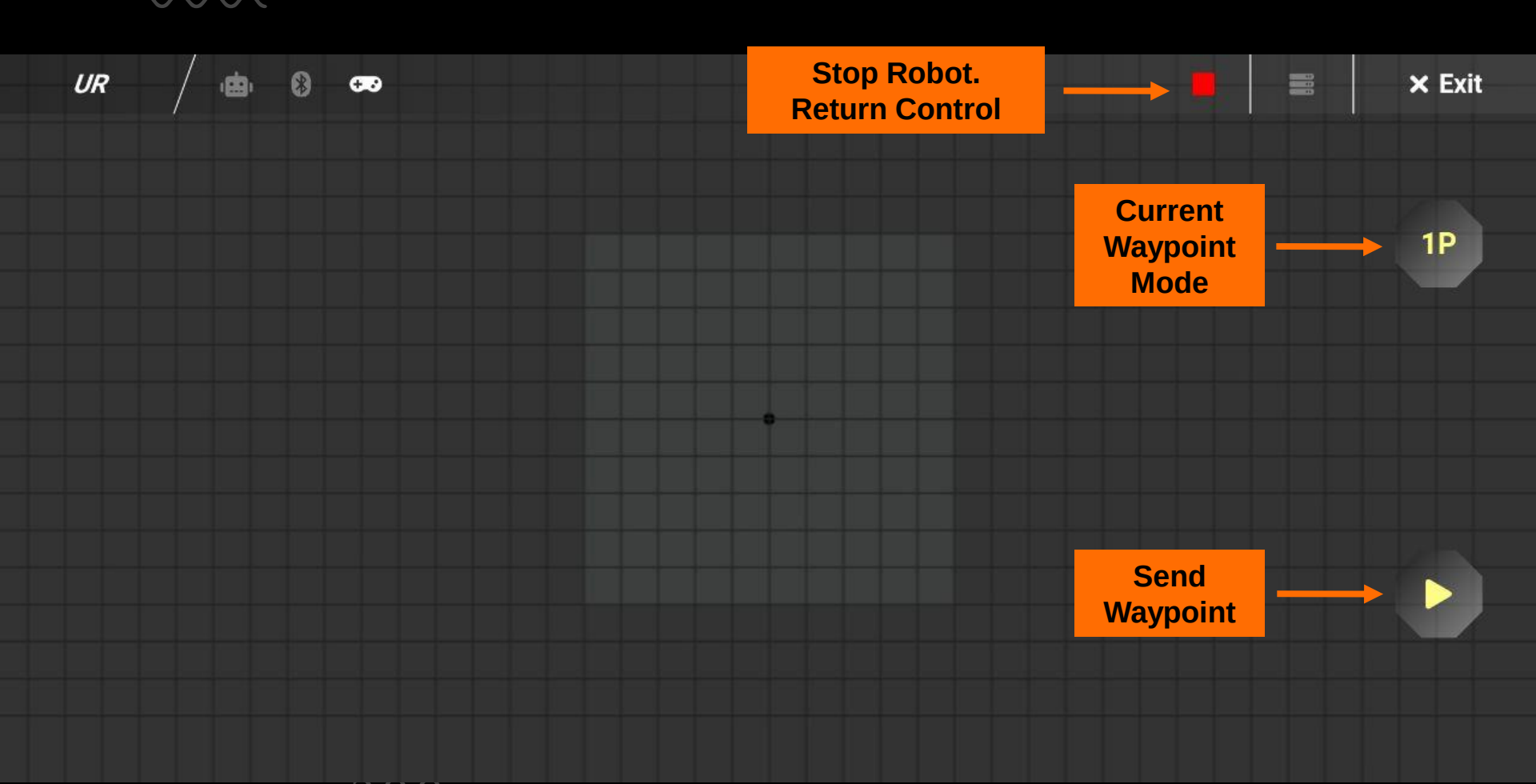
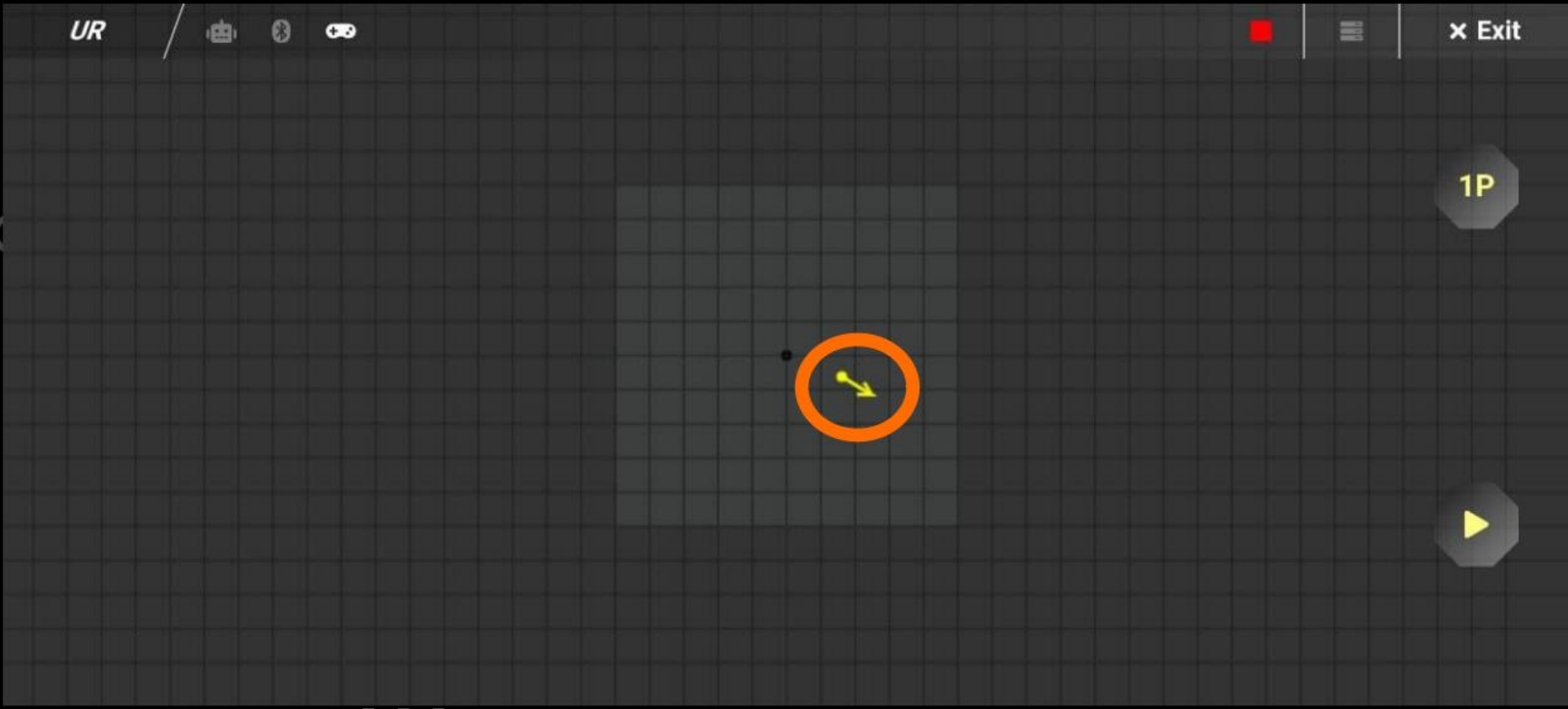
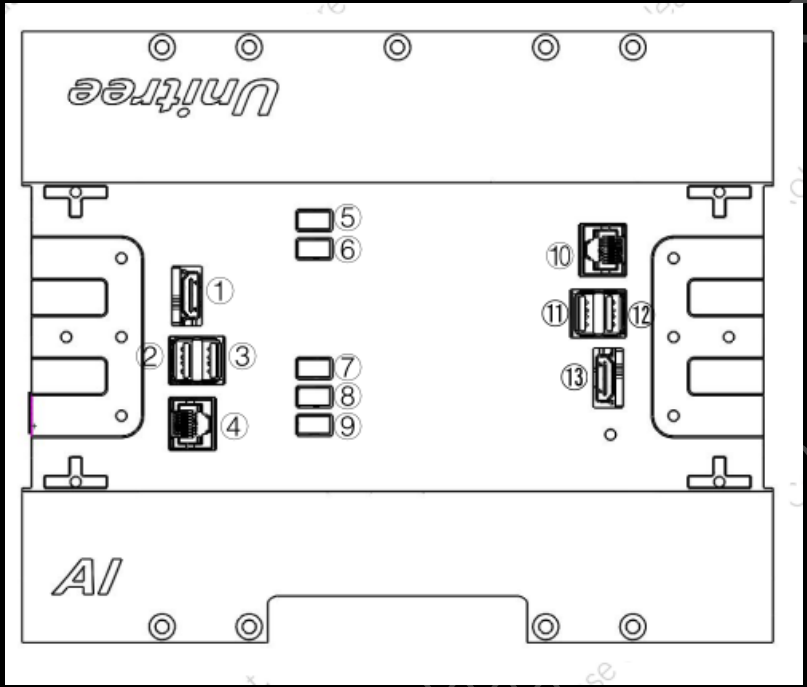

UR
Stop Robot.
Return Control
Exit
Current
Waypoint
Mode
1P
Send
Waypoint



Weston Robot

Internal Components of A1

- Low-level Controller
  - Walking Mode
- Upboard
  - Sports Mode
- Raspberry Pi / NX
  - Video stream
  - Wi-Fi Access point
- Ethernet Switch
  - Connects 3 main components
  - Addition of computer



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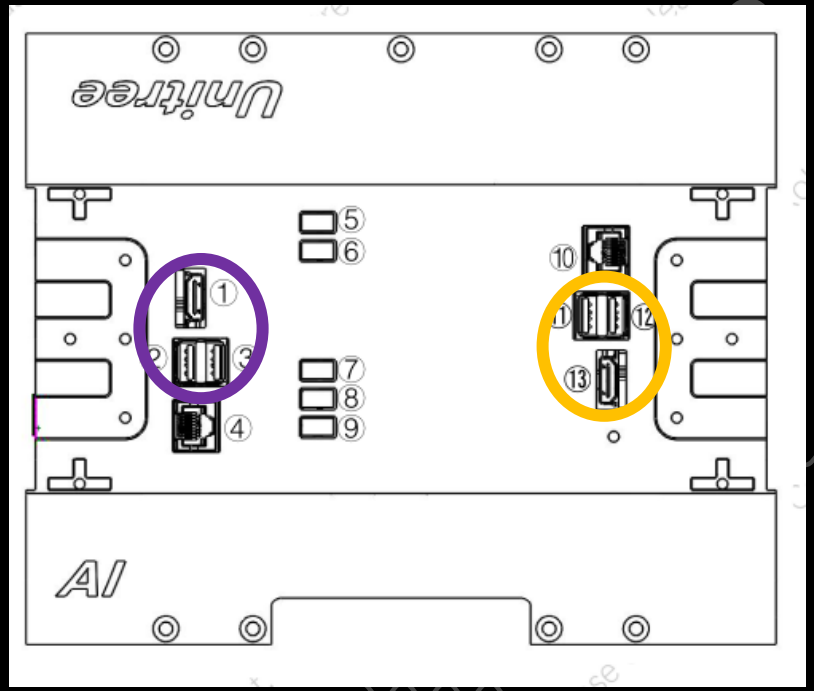
Access Raspi/NX / Upboard

- Hardware
  - HDMI Monitor
  - Mouse
  - Keyboard

- Network (192.168.123.X)
  - 192.168.123.161 (Upboard)
  - 192.168.123.12 (Raspi / NX)
  - 192.168.123.10 (Low-level Controller)

- Caution
  - Place legs facing up before boot
  - Stand away from the Robot Dog when boot
  - Legs might Spasm

- Other Peripherals
  - 192.168.11.1 (2D LiDAR)



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Connect to Raspi

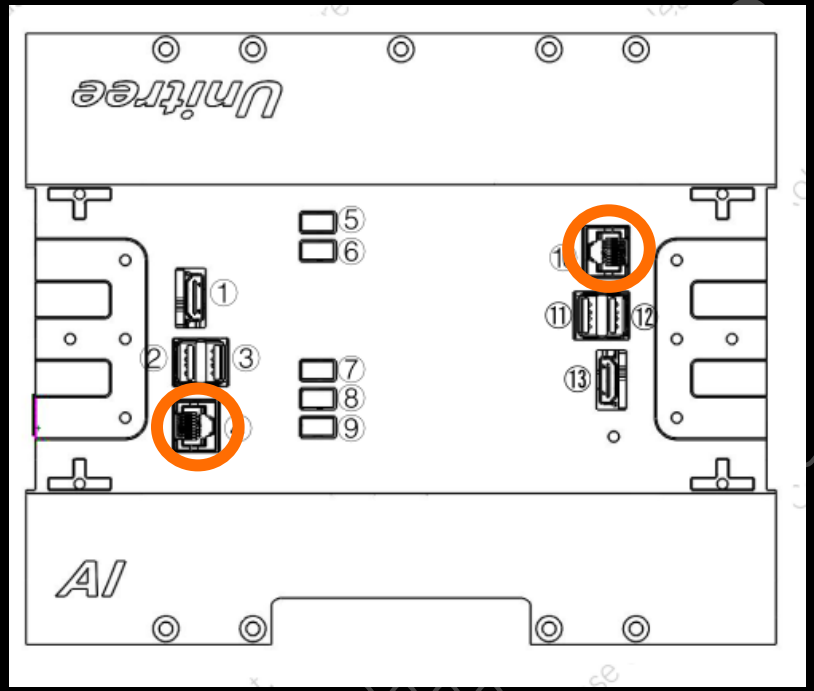
- Wi-Fi
  - UnitreeRoboticsA1-XXX
  - Password: 00000000

- Log in
  - IP Address: 192.168.123.12 (External Network)
  - Username: pi (raspberry pi), unitree (nx)
  - Password: 123

- Example:
  - ssh pi@192.168.123.12
  - Password: 123

Add External Computer

- Connect to ethernet switch
- ifconfig
- ping
  - 192.168.123.161 (Upboard)
  - 192.168.123.12 (Raspi/Nx)
  - 192.168.123.10 (Low-level Controller)



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Questions?